

RESEARCH METHODS IN COMPUTING

SCS301

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Meaning of Research

- ▶ Research may be very broadly defined as systematic gathering of data and information and its analysis for advancement of knowledge in any subject.
- ▶ Research is carrying out a diligent inquiry or a critical examination of a given phenomenon.
- ▶ Research attempts to find answer intellectual and practical questions through application of systematic methods
- ▶ Research is, thus, an original contribution to the existing stock of knowledge making for its advancement.
- ▶ It is the pursuit of truth with the help of study, observation, comparison and experiment.
- ▶ The search for knowledge through objective and systematic method of finding solution to a problem is research.
- ▶ The purpose of research is to discover answers to questions through the application of scientific procedures.
- ▶ The main aim of research is to find out the truth which is hidden and which has not been discovered as yet.

Purpose/Objective of Research

- ▶ **Exploration:** To discover or explore **new knowledge**
- ▶ **Description:** To **describe** or **gain familiarity with a phenomenon** or the **characteristics** of phenomena
- ▶ **Explanation:** To **achieve new insights into it** (studies with this object in view are termed as exploratory or formulative research studies);
- ▶ **Prediction:** To predict future occurrences or behaviours.
- ▶ **Application:** To use findings for practical applications
- ▶ **To portray accurately the characteristics** of a particular individual, situation or a group (studies with this object in view are known as descriptive research studies);
- ▶ To **determine the frequency** with which something occurs or with which it is associated with something else (studies with this object in view are known as diagnostic research studies);
- ▶ To **test a hypothesis of a causal relationship between variables** (such studies are known as hypothesis-testing research studies).

Motivation for Research Study

The possible motives for doing research may be either one or more of the following:

1. **Academic Pursuit:** Desire to get a **research degree** along with its consequential benefits;
2. **Personal Satisfaction:** The desire to gain knowledge for personal growth
3. **Professional Advancement:** To achieve recognition, promotions, and career growth.
4. **Intellectual joy:** Desire to get **intellectual joy** of doing some creative work;
5. **Social Impact:** aim to solve **societal issues and contribute to the community**.
6. **Curiosity:** A natural inclination to learn and discover new things and Desire to **face the challenges in solving unsolved problems**
7. **Respect:** Desire to get **respectability**.

► However, this is not an exhaustive list of factors motivating people to undertake research studies.

► Many more factors such as

1. **directive of government or research organisation or university**
2. **employment conditions**
3. **desire to understand causal relationships**
4. **social thinking and awakening**

Definition of basic terms used in research

- ▶ **Population:** it refers to an entire group of individuals, events or objects having a common observable characteristic.
- ▶ **Sample:** It is a smaller group obtained from the accessible population.
- ▶ **Sampling:** It is the process of selecting a number of individuals for a study in such a way that the individuals selected represent the population.
- ▶ **Variable:** It is a measurable characteristic that assumes different values among the subjects. They can be dependent, independent, intervening, confounding or antecedent variables.
- ▶ **Data:** refers to all information a researcher gathers for his or her study. Can be secondary data or primary data.
- ▶ **Parameter:** It is a characteristic that is measurable and can assume different values in the population.
- ▶ **Statistics:** it is the science of organizing, describing and analyzing data. Descriptive and inferential statistics.
- ▶ **Objective:** it refers to the specific aspects of the phenomenon under study that the researcher desires to bring out at the end of the research study.
- ▶ **Literature review:** It involves locating, reading and evaluating reports of previous studies, observations and opinions related to the planned study.
- ▶ **Hypothesis:** It is a researcher's anticipated explanation or opinion regarding the result of the study.
- ▶ **Theory:** It is a set of concepts or constructs and the interrelations that are assumed to exist among those concepts. It provides the basis for establishing the hypothesis to be tested in the study.
- ▶ **A construct** is an image or idea specifically invented for a given research and/or theory-building purpose
- ▶ **A concept** is a bundle of meanings or characteristics associated with certain events, objects, conditions, situations, and behaviors. Concepts have been developed over time through shared usage

Sources of Knowledge

- ▶ **Research**
- ▶ **Experience**: Empiricists attempt to describe, explain, and make predictions through observation.
- ▶ **Tradition**: Rationalists believe all knowledge can be deduced from known laws or basic truths of nature
- ▶ **Authority**: They serve as important sources of knowledge, but should be judged on integrity and willingness to present a balanced case.
- ▶ **Intuition**: it is the perception, explanation or insight into phenomena by instinct.

TYPES OF RESEARCH

Types of research can be classified in many different ways. some major ways of classifying research include the following.

1. **Descriptive Research:** It describes the **characteristics of a phenomenon or population** and **focuses on finding facts** to determine its nature as it exists.
2. **Analytical Research:** Analyzes information and data to **understand relationships and causality**, and is focused on determining the **validity of hypotheses** based on the analysis of **collected facts**.
3. **Applied Research:** This research is conducted to address **practical problems** and assist in decision-making across various fields, including product design, process design, and policy making.
4. **Fundamental research** The research is conducted more to **satisfy intellectual** curiosity than to use the findings for any immediate practical application..
5. **Quantitative research** Analyzes, interprets, and resolves issues using **numerical data and statistical analysis** to **draw inferences**
6. **Qualitative research** Qualitative research based on words, feelings, emotions, sounds, and other non-numerical, unquantifiable elements.
7. **Basic research/fundamental/ Conceptual/:** Conceptual research focuses on **abstract ideas** and **fundamental principles**. It seeks to enhance knowledge through the **development of new concepts and theories without immediate practical application**.
8. **Empirical Research:** A method of acquiring knowledge through **observation and experience**. It emphasizes **real-world data** and **systematically** gathered evidence, enhancing the **reliability and credibility** of conclusions.

Research Approaches

These research approaches offer different perspectives and tools for investigating questions and solving problems

1. Qualitative Research

- ▶ Qualitative Research is **primarily exploratory research**.
- ▶ **Focus:** Understanding **phenomena** from a holistic perspective.
- ▶ **Data:** **Non-numerical**, such as interviews, observations, and textual analysis.
- ▶ **Purpose:** To **explore** and **gain insights** into complex issues, understanding the "why" and "how" behind **behaviors** and **experiences**.
- ▶ **Example Methods:** Case studies, ethnography, narrative analysis.
- ▶ The **sample size** is typically small, and **respondents are selected** to fulfil a given quota.

2. Quantitative Research

- ▶ Quantitative research seeks to **quantify** problems by generating **numerical data** that can be **analyzed statistically**.
- ▶ **Focus:** Collection and analysis of numerical data and **generalize results** from a larger sample population.
- ▶ **Data:** **Numerical**, such as surveys, experiments, and statistical analysis.
- ▶ **Purpose:** To **quantify variables** and understand patterns, relationships, and causality.
- ▶ **Example Methods:** Surveys, controlled experiments, longitudinal studies.

3. Mixed-Methods Approach:

- ▶ Combines **qualitative** and **quantitative** methods to provide a comprehensive understanding by **leveraging the strengths of both approaches**.

Significance of Research

Knowledge Advancement:

Contribution: Enhances and expands the existing body of knowledge.

Impact: Provides new insights, theories, and understanding in various fields.

Problem-Solving:

Objective: Addresses and finds solutions to specific issues.

Impact: Improves processes, technologies, and practices to resolve problems effectively.

Policy Making:

Role: Informs and shapes public policies based on empirical evidence.

Impact: Ensures that policies are effective, relevant, and beneficial for society.

Innovation:

Driver: Propels technological and methodological advancements.

Impact: Leads to developing new tools, techniques, and technologies.

Societal Development:

Benefit: Improves societal well-being and quality of life.

Impact: Contributes to economic growth, healthcare, education, and human development.

Economic Growth

Role: Promote economic advancements by fostering innovation and improving industrial processes.

Impact: Contributes to the development of new markets, products, and services, boosting the economy.

Education Enhancement

Role: Education research can lead to better teaching methods, curricula, and learning tools.

Impact: Enhances the quality of education and learning outcomes for students.

Health Improvements

Role: Medical and health research can lead to new treatments, drugs, and healthcare practices.

Impact: Improves public health and extends life expectancy.

Environmental Sustainability

Role: Research in environmental sciences can identify sustainable practices and solutions to environmental problems.

Impact: Promotes conservation and sustainable use of resources, protecting ecosystems.

Technological Advancement

Role: Research drives technological breakthroughs and advancements.

Impact: Leads to the development of cutting-edge technologies that improve efficiency and quality of life.

Cultural and Social Understanding

Role: Research in social sciences and humanities enhances our understanding of cultural, social, and historical contexts.

Impact: Promotes cultural appreciation, social cohesion, and informed decision-making.

Industry Standards and Best Practices

Role: Research helps in developing standards and best practices across various industries.

Impact: Ensures quality, safety, and efficiency in industrial processes and products.

Personal Development

Role: Conducting research fosters critical thinking, problem-solving, and analytical skills in individuals.

Impact: Contributes to personal and professional growth.

Criteria of a Good Research

Systematic:

Definition: Follows a **structured and organized** approach.

Importance: Ensures that the research process is **methodical and consistent**.

Logical:

Definition: Based on **sound reasoning and logical analysis**.

Importance: Guarantees that the conclusions drawn are **rational and well-founded**.

Empirical:

Definition: Grounded in **observable and measurable evidence**.

Importance: Provides concrete data to **support findings, making them reliable and valid**.

Replicable:

Definition: Can be **repeated and verified** by other researchers.

Importance: Allows for the **validation** of results through repeated studies, ensuring **accuracy and reliability**.

Ethical:

Definition: Adheres to ethical **standards and principles**.

Importance: Ensures the **integrity** of the research **process** and **protects the rights and well-being of participants**.

Research Method vs Research Methodology

Research methods

- ▶ **Research methods** may be understood as all those **methods/ techniques** that are used for conduction of research.
- ▶ **Techniques** and **tools** used for data collection and analysis (e.g., surveys, experiments).
- ▶ Research methods or techniques*, thus, refer to **the methods the researchers use in performing research operations.**

Research Methodology

- ▶ Research methodology is a way to **systematically solve the research problem.**
- ▶ It may be understood as **a science of studying how research is done scientifically.**
- ▶ The overall **strategy** and **rationale** guiding the research process, including the philosophical underpinnings and choice of methods.

Integration of Research and the Scientific Method

1. Empirical Evidence

- ▶ **Description:** Relies on data and observations that can be verified.
- ▶ **Importance:** Ensures that findings are based on concrete, observable evidence.

2. Relevant Concepts

- ▶ **Description:** Utilizes concepts that are pertinent to the research question.
- ▶ **Importance:** Ensures that the research is focused and relevant to the topic.

3. Objective Considerations

- ▶ **Description:** Maintains objectivity throughout the research process.
- ▶ **Importance:** Avoids bias and ensures that the research findings are neutral and accurate.

4. Ethical Neutrality

- ▶ **Description:** Ensures statements are accurate and unbiased.

- ▶ **Importance:** Adheres to ethical standards, maintaining the integrity of the research.

5. Probabilistic Predictions

- ▶ **Description:** Provides predictions based on probability.
- ▶ **Importance:** Allows for uncertainty and variability in research findings, making them more realistic.

6. Transparency and Replicability

- ▶ **Description:** Methods are open to scrutiny and can be replicated.
- ▶ **Importance:** Ensures that other researchers can verify and validate the findings.

7. General Axioms

- ▶ **Description:** Aims to formulate broad, scientific theories.
- ▶ **Importance:** Contributes to the development of universal principles and theories in science.

IMPORTANCE OF KNOWING HOW RESEARCH IS DONE

Critical Thinking

- ▶ Enhancement: Improves **analytical** and **critical thinking** skills.
- ▶ Impact: Allows individuals to **assess information** rigorously and make **well-founded judgments**.

Informed Decisions

- ▶ Capability: Enables **making decisions based on evidence and sound reasoning**.
- ▶ Impact: Leads to more **accurate, effective, and reliable** outcomes in various situations.

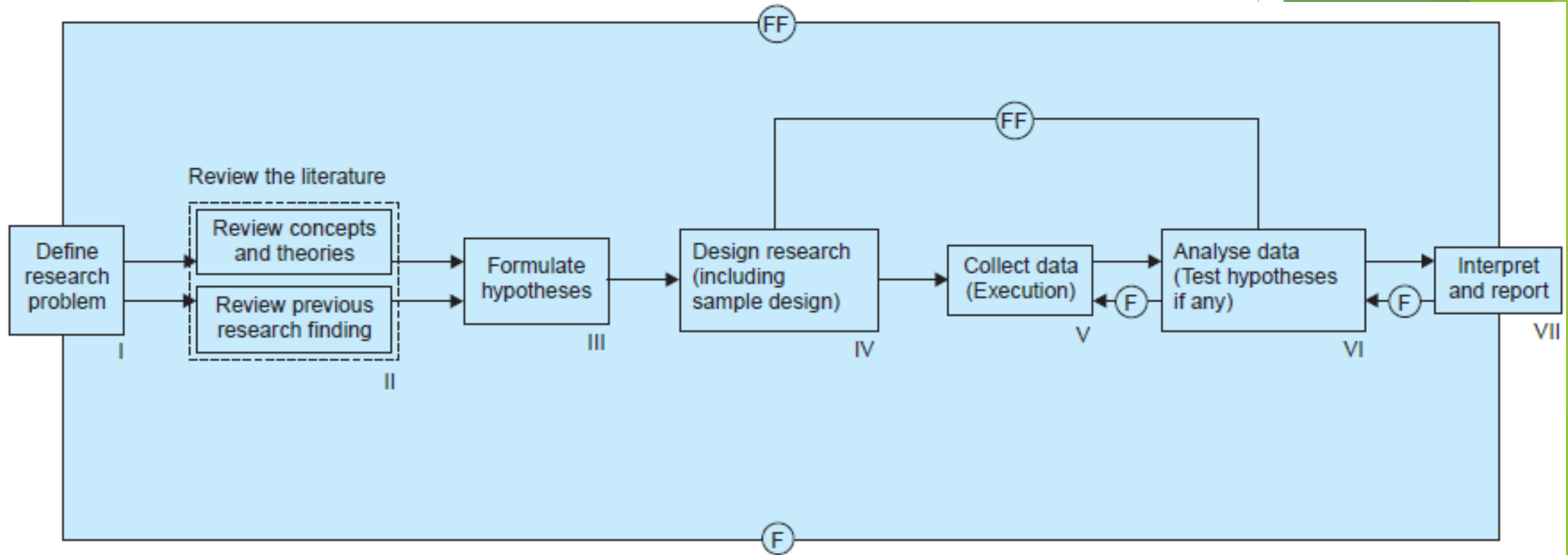
Credibility

- ▶ Building: Establishes **credibility and trustworthiness** in the research findings.
- ▶ Impact: Gains confidence from **peers, stakeholders, and the wider community**.

Innovation and Improvement

- ▶ Encouragement: Promotes **innovation and continuous improvement** across different fields.
- ▶ Impact: Drives progress, leading to **new technologies, methodologies, and enhanced practices**.

Research Process



Where (F) = feed back (Helps in controlling the sub-system to which it is transmitted)

(FF) = feed forward (Serves the vital function of providing criteria for evaluation)

Research Process in Research Methodology

- ▶ Step 1: Define the Problem: Clearly state the research problem or question.
- ▶ Step 2: Review Literature: Conduct a comprehensive review of existing literature.
- ▶ Step 3: Formulate Hypotheses: Develop hypotheses or research questions.
- ▶ Step 4: Design Research: Choose the research design and methodology.
- ▶ Step 5: Collect Data: Gather data using appropriate methods.
- ▶ Step 6: Analyze Data: Analyze the collected data using statistical or qualitative techniques.
- ▶ Step 7: Interpret Results: Interpret the findings and draw conclusions.
- ▶ Step 8: Report Findings: Document and disseminate the research findings.

Research Process in Research Methodology: Detailed

1. Define the Problem:

Objective: Clearly state the research problem or question.

- ▶ Start by identifying the specific issue or area of interest.
- ▶ Formulate a clear and concise research question or problem statement.
- ▶ Ensure the problem is researchable and significant to the field of study.
- ▶ Example: "What are the effects of remote work on employee productivity?"

2. Review Literature:

Objective: Conduct a comprehensive review of existing literature.

- ▶ Gather existing research and studies related to the research problem.
- ▶ Identify gaps, inconsistencies, and areas that require further investigation.
- ▶ Use databases, journals, books, and credible sources.
- ▶ Summarize key findings and highlight their relevance to your research.
- ▶ Example: Reviewing studies on remote work, productivity, and related factors.

3. Formulate Hypotheses:

Objective: Develop hypotheses or research questions.

- ▶ Based on the literature review, propose tentative explanations or predictions.
- ▶ Hypotheses should be specific, testable, and related to the research problem.
- ▶ Example: "Remote work increases employee productivity due to fewer distractions."

4. Design Research:

Objective: Choose the research design and methodology.

- ▶ Select the appropriate research design (e.g., experimental, survey, case study).
- ▶ Decide on the methods for data collection (e.g., interviews, questionnaires, observations).
- ▶ Determine the sampling method and sample size.
- ▶ Develop a detailed research plan outlining procedures and protocols.
- ▶ Example: Using a survey to collect data from remote workers.

5. Collect Data:

Research Process in Research Methodology: Detailed

Objective: Gather data using appropriate methods.

- ▶ Implement the data collection plan as outlined in the research design.
- ▶ Ensure data is collected systematically and ethically.
- ▶ Use tools and techniques suitable for the research (e.g., online surveys, field observations).
- ▶ Example: Distributing an online survey to remote employees to gather responses.

6. **Analyze Data:**

Objective: Analyze the collected data using statistical or qualitative techniques.

- ▶ Organize and clean the collected data for analysis.
- ▶ Use statistical methods (e.g., regression analysis, ANOVA) for quantitative data.
- ▶ Apply qualitative methods (e.g., thematic analysis, coding) for qualitative data.
- ▶ Interpret the data to identify patterns, relationships, and trends.
- ▶ Example: Analyzing survey responses to determine the impact of remote work on productivity.

7. **Interpret Results:**

Objective: Interpret the findings and draw conclusions.

- ▶ Relate the analysis results to the research questions or hypotheses.
- ▶ Discuss the implications of the findings in the context of existing literature.
- ▶ Identify limitations and suggest areas for future research.
- ▶ Example: Concluding that remote work positively affects productivity, with recommendations for employers.

8. **Report Findings:**

Objective: Document and disseminate the research findings.

- ▶ Details:
- ▶ Prepare a comprehensive report or paper detailing the research process, findings, and conclusions.
- ▶ Include sections such as introduction, methodology, results, discussion, and conclusion.
- ▶ Share the findings through presentations, publications, or online platforms.
- ▶ Example: Publishing the research in a journal or presenting at a conference.

Research Problem

Selecting a Research Problem: Points to Consider:

1. **Relevance:** Ensure the problem is significant and relevant to the field.
2. **Feasibility:** Verify that the problem can be investigated with available resources and within time constraints.
3. **Novelty:** Identify gaps in existing research and focus on unexplored or underexplored areas.
4. **Interest:** Choose a problem that genuinely interests you to maintain motivation.
5. **Clarity:** Make sure the problem is clear and well-defined.

Necessity of Defining a Research Problem:

- **Focus:** Provides a clear direction for the research.
- **Guidance:** Helps in formulating research questions and hypotheses.
- **Scope:** Sets the boundaries for the study, ensuring it remains manageable.
- **Relevance:** Ensures the research addresses significant and relevant issues.
- **Evaluation:** Allows for the assessment of research success and relevance.

Techniques Involved in Defining a Research Problem

Literature Review:

- ▶ **Purpose:** Identify existing knowledge, gaps, and areas needing further exploration.
- ▶ **Approach:** Review academic journals, books, and credible sources related to the topic.
- ▶ **Outcome:** Gain insights into previous studies and refine the research problem based on identified gaps.

Expert Consultation:

- ▶ **Purpose:** Leverage the experience and knowledge of experts in the field.
- ▶ **Approach:** Seek advice and feedback from professors, researchers, and industry professionals.
- ▶ **Outcome:** Obtain valuable perspectives and suggestions to refine the research problem.

Preliminary Research:

- ▶ **Purpose:** Conduct initial studies to explore potential research areas.
- ▶ **Approach:** Use surveys, interviews, or small-scale experiments to gather preliminary data.
- ▶ **Outcome:** Identify feasible and significant research problems based on initial findings.

Brainstorming:

- ▶ **Purpose:** Generate a wide range of ideas and potential research problems.
- ▶ **Approach:** Engage in brainstorming sessions with peers, mentors, or research

teams.

- ▶ **Outcome:** Create a list of possible research problems and narrow down based on criteria such as relevance and feasibility.

Problem Identification Matrix:

- ▶ **Purpose:** Compare and evaluate different research problems systematically.
- ▶ **Approach:** Use a matrix to score potential problems based on criteria such as significance, feasibility, and researchability.
- ▶ **Outcome:** Select the most suitable research problem through a structured evaluation process.

Gap Analysis:

- ▶ **Purpose:** Identify gaps between current knowledge and desired outcomes.
- ▶ **Approach:** Analyze existing literature and research to determine what has not yet been addressed.
- ▶ **Outcome:** Define a research problem that aims to fill identified gaps.

SWOT Analysis:

- ▶ **Purpose:** Evaluate strengths, weaknesses, opportunities, and threats related to potential research problems.
- ▶ **Approach:** Conduct a SWOT analysis to assess the viability of different research topics.
- ▶ **Outcome:** Select a research problem with strong potential and minimal risks.

Characteristics of a good problem statement

Clarity:

- ▶ **Definition:** Clearly and concisely states the problem.
- ▶ **Importance:** Ensures that the problem is understandable and free from ambiguity, making it easier to address and resolve.

Specificity:

- ▶ **Definition:** Focuses on a specific issue or question.
- ▶ **Importance:** Narrows down the scope of the research to a manageable and focused area, facilitating in-depth investigation.

Relevance:

- ▶ **Definition:** Addresses a significant issue in the field.
- ▶ **Importance:** Ensures that the problem is worth investigating and contributes to the advancement of knowledge or practice in the field.

Researchability:

- ▶ **Definition:** Can be investigated through empirical methods.
- ▶ **Importance:** Confirms that the problem can be studied and analyzed using scientific methods, providing reliable and valid results.

Feasibility:

- ▶ **Definition:** Can be addressed with available resources and within time constraints.
- ▶ **Importance:** Ensures that the research problem can be realistically tackled given the resources, time, and capabilities at the researcher's disposal.

Stating the Objectives and Research Questions

Objective:

- ▶ **Definition:** Clearly define what the research aims to achieve.
- ▶ **Importance:** Provides a clear roadmap for the research process, ensuring that all efforts are aligned with the desired outcomes.

Importance:

- ▶ **Guidance:** Offers direction and focus for the entire research project.
- ▶ **Measurement:** Helps in assessing the success and relevance of the research.
- ▶ **Clarity:** Ensures that all stakeholders understand the purpose and goals of the study.

Types of Objectives:

1. General Objectives:

- ▶ **Definition:** Broad goals that provide overall direction.
- ▶ **Purpose:** Establishes the primary aim of the research.
- ▶ **Example:** "To investigate the impact of remote work on employee productivity."

2. Specific Objectives:

- ▶ **Definition:** Narrow, precise goals that detail the steps to achieve the general objectives.
- ▶ **Purpose:** Breaks down the general objectives into actionable and measurable tasks.

1. Examples:

1. "To assess changes in productivity levels among remote workers over six months."
2. "To identify factors that influence remote work productivity."
3. "To compare productivity between remote and on-site employees."

Research Questions:

- **Definition:** Clear, focused, and concise queries that guide the direction and scope of a research study.
- **Importance:** Sets the foundation for the investigation and determines what the researcher aims to discover or understand.

▶ Examples:

1. Descriptive Research Questions:

1. "What are the demographic characteristics of remote workers?"

2. Comparative Research Questions:

1. "How does remote work productivity compare to on-site work productivity?"

3. Correlational Research Questions:

1. "Is there a correlation between remote work and job satisfaction?"

4. Causal Research Questions:

1. "Does remote work cause an increase in employee productivity?"

FORMULATING HYPOTHESES

A hypothesis is a tentative explanation or prediction about the relationship between two or more variables that can be tested through research.

Steps to Formulate a Hypothesis:

1. Identify the Research Problem: OR question

- ▶ Start with a well-defined research problem or question.
- ▶ Example: "How do different **software metrics** impact **software quality** and **maintenance** in agile development environments?"

2. Conduct a Literature Review:

- ▶ Review existing research to understand current knowledge on software metrics and their effects on software quality.
- ▶ Example: Studies have indicated that metrics like cyclomatic complexity and code churn can influence software maintainability and defect rates.

3. Define Variables:

- ▶ Identify the independent (predictor) and dependent (outcome) variables.
- ▶ Independent Variables: Different software metrics (e.g., cyclomatic complexity, code churn, code coverage).
- ▶ Dependent Variables: Software quality measures (e.g., defect density, maintainability index).

4. Formulate the Hypothesis:

1. Develop a clear and concise statement predicting the relationship between the variables.
2. Example: "Higher cyclomatic complexity in software code is associated with an increase in defect density in agile development environments."

5. Ensure Testability:

1. Make sure the hypothesis is specific and can be tested empirically.
2. Example: The hypothesis can be tested by analyzing defect density in codebases with varying levels of cyclomatic complexity.

6. Refine and Clarify:

1. Refine the hypothesis to ensure clarity and precision.
2. Example: "Software modules with higher cyclomatic complexity will exhibit higher defect density compared to modules with lower cyclomatic complexity in agile projects."

Characteristics of a Good Hypothesis:

1. **Testability:** Can be empirically tested and validated through research.
2. **Clarity:** Clearly and precisely stated, avoiding ambiguity.
3. **Relevance:** Directly related to the research problem and objectives.
4. **Specificity:** **Focuses** on a specific aspect of the research problem.
5. **Simplicity:** Simple, avoiding unnecessary complexity.
6. **Predictive Power:** Makes a specific prediction about the relationship between variables.

Examples:

1. **Descriptive Hypothesis:** "Code churn levels in agile projects are higher during the initial development phases compared to maintenance phases."
2. **Comparative Hypothesis:** "Software modules with higher cyclomatic complexity will exhibit higher defect density compared to modules with lower cyclomatic complexity in agile projects."
3. **Correlational Hypothesis:** "There is a positive correlation between code coverage and software maintainability in agile development environments."
4. **Causal Hypothesis:** "Higher code churn in agile development projects causes an increase in defect density due to frequent changes in the codebase."

TYPES OF VARIABLES

- ▶ A variable is a **measurable characteristic** that assumes different values among the subjects.
- ▶ **Independent variables / Predictor variables**
- ▶ It is a variable that a **researcher manipulates** in order to determine its effect or influence on another variable. They predict the amount of variation that occurs in another variables.
- ▶ **Dependent variables / criterion variables**
- ▶ It is the variable that is measured, predicted or monitored and is expected to be affected by **manipulation** of an independent variable. They attempt to indicate the total influence arising from the effects of the independent variable.
- ▶ **Extraneous variables**
- ▶ They are those variables that affect the outcome of a research study **either because the researcher is not aware of their existence** or if the researcher is aware, she or he has no control over them.
- ▶ **Control variables / concomitant / covariate or blocking variables**
- ▶ They are extraneous variables that are **built into the study**. Extraneous variables are variables, which influence the results of a study when they are not controlled.
- ▶ Reasons for introducing control variables: It increases the **validity of the data** and It leads to more **convincing generalizations**.
- ▶ **Intervening variables**
- ▶ An **intervening variable** is a hypothetical internal state that is used to explain relationships between observed variables, such as independent and dependent variables, in **empirical research**. With an extraneous variable, there is no causal link between the independent and dependent variable, but they are independently associated with a third variable

LITERATURE REVIEW

- ▶ The review of literature involves the **systematic identification, location and analysis of documents containing information related to the research problem being investigated**. Avoid unnecessary and unintentional duplication
- ▶ It should be **extensive and thorough** because it is aimed at obtaining **detailed knowledge of the topic** being studied.
- ▶ **Determines** what has already been done related to the research problem being studied.
- ▶ Helps reveal the **strategies, procedures and measuring instruments** that have been found useful in investigating the problem in question.
- ▶ It helps the researcher to **limit the research problem** and to define it better.
- ▶ Helps to **determine new approaches** and stimulates new ideas.
- ▶ **Specific suggestions and recommendations** for further research can be found by reviewing literature.
- ▶ Helps **to reveal gaps in information** and areas where major questions still remain.

Sources of literature

- ▶ **Primary sources:** are direct descriptions of any occurrence by an individual who actually observed or witnessed the occurrence.
- ▶ **Secondary source:** they include any publications written by an author who was not a direct observer or participant in the events described.
- ▶ **Examples**
 - ▶ Scholarly journals
 - ▶ Theses and dissertations
 - ▶ Government documents
 - ▶ Papers presented at conferences
 - ▶ Books
 - ▶ References quoted in books
 - ▶ International indices
 - ▶ Abstracts
 - ▶ Periodicals
 - ▶ The Africana section of the library
 - ▶ Reference section of the library
 - ▶ Grey literature
 - ▶ Inter-library loan
 - ▶ The British lending library
 - ▶ The internet
 - ▶ Microfilm

What is the meaning of Research design ?

- ▶ The research design is a **comprehensive master plan** of the research study to be undertaken, giving a **general statement of the methods** to be used.
- ▶ The **plan and structure of investigation** so conceived as to obtain answers to research questions.
- ▶ The function of a research design is to ensure that requisite **data in accordance with the problem** at hand is collected **accurately and economically**.
- ▶ it is the framework, a blueprint for the research study which specifies the methods and procedures for the collection, measurement and analysis of data.

MAJOR TYPES OF RESEARCH DESIGN

Exploratory studies

- Exploration is particularly useful when researchers lack a clear idea of the problems they will meet during the study.

Descriptive Studies

- It is the process of collecting data in order to test hypotheses or to answer questions concerning the current status of the subjects in the study. It determines and reports the way things are.
- Provides answers to questions like Who? What? When? Where? How? It attempts to describe such things as possible behaviour, attitudes, values and characteristics.

Causal Research

- It is used to explore relationships between variables. It determines reasons or causes for the current status of the phenomenon under study.

Correlation Methods

- It describes in quantitative terms the degree to which variables are related. It explores relationships between variables and also tries to predict a subject's score on one variable given his or her score on another variable.

THE SAMPLE DESIGN

Definition: Sample design refers to the **framework or plan** used to select a **sample** from a **larger population**.

It outlines the **method** and **process** for choosing the **sample**, ensuring that it is **representative** of the population and **suitable** for the research objectives.

Key Components of Sample Design:

1. **Population:** The **entire group** of individuals or elements that the researcher is interested in studying.
2. **Sampling Frame:** A list or **database containing all members** of the population from which the sample will be drawn.
3. **Sample Size:**
 - ▶ The **number of individuals or elements selected** from the population to participate in the study.
 - ▶ Factors influencing sample size include **the research objectives, desired precision, and available resources**.
4. **Sampling Method:** The technique used to **select the sample**.
5. This can be broadly categorized into **probability sampling methods and non-probability sampling methods**.

Steps in Sample Design:

1. **Define the Population:** Identify the **target population** for the study.
2. **Develop the Sampling Frame:** Create a comprehensive list or database of all members of the population.

3. **Determine the Sample Size:** Calculate the appropriate **sample size** based on research objectives and constraints.
4. **Select the Sampling Method:** Choose the most suitable **sampling method** based on the research goals and characteristics of the population.
5. **Implement the Sampling Plan:** Execute the chosen sampling method to select the sample.
6. **Verify the Sample:** Ensure the selected sample is representative and free from bias.

Importance of Sample Design:

- ▶ **Accuracy:** A well-designed sample ensures accurate and reliable results.
- ▶ **Representativeness:** Ensures that the sample reflects the characteristics of the population.
- ▶ **Generalizability:** Allows findings to be generalized to the entire population.
- ▶ **Efficiency:** Optimizes resources and minimizes cost and time.
- ▶ **Bias Reduction:** Reduces sampling bias and enhances the validity of the study.

Probability Sampling Methods

- ▶ Probability sampling methods ensure that every member of a population has a known, or equal, chance of being selected for a sample.
- ▶ This statistical approach ensures that the sample is representative of the population, which allows researchers to make accurate and reliable inferences and generalizations about the entire population.

Types of Probability Sampling Methods

1. Simple Random Sampling:

- **Description:** Every member of the population has an equal chance of being selected.
- **Procedure:** Use random number generators or drawing lots to select participants.
- **Example:** Selecting 100 employees from a company of 1000 employees using a random number generator.

2. Systematic Random Sampling:

- ▶ **Description:** Selects participants at regular intervals from an ordered list.
- ▶ **Procedure:** Determine the sampling interval (e.g., every 5th person) and start at a random point.
- ▶ **Example:** Surveying every 10th customer who enters a store.

- ▶ A random starting point is selected and then every k^{th} member of the population is selected for the sample.

3. Stratified Random Sampling:

- ▶ **Description:** Divides the population into subgroups (strata) based on specific characteristics, then randomly samples from each subgroup.
- ▶ **Procedure:** Identify strata, then use simple random sampling within each stratum.
- ▶ **Example:** Sampling students from different academic departments to ensure representation from each department.

4. Cluster Sampling:

- ▶ **Description:** Divides the population into clusters, randomly selects clusters, then samples all or a subset of individuals within selected clusters.
- ▶ **Procedure:** Identify clusters, randomly select clusters, then sample within clusters.
- ▶ **Example:** Surveying households in randomly selected neighborhoods within a city

- ▶ **5. Multi-Stage Sampling:** Combines two or more probability sampling methods in stages.

Non - Probability Sampling Methods

- ▶ Non-probability sampling methods are techniques in which **samples are selected based on subjective judgment** instead of **random selection**.
- ▶ In these methods, **not every member of the population has an equal or known chance** of being included in the sample.
- ▶ These methods are often used when **probability sampling is impractical or impossible**.

Types of non-probability sampling methods:

1. Convenience Sampling:

- ▶ **Description:** Participants are selected based on their availability and ease of access.
- ▶ **Example:** Surveying people at a shopping mall because they are easy to reach.

2. Judgmental (Purposive) Sampling:

- ▶ **Description:** Participants are selected based on the researcher's judgment and knowledge of the population.
- ▶ **Example:** Selecting experts in a particular field for an in-depth study.

3. Quota Sampling:

- ▶ **Description:** Ensures that specific subgroups are represented

in the sample by setting quotas.

- ▶ **Example:** Ensuring that a sample includes a certain number of participants from different age groups.

4. Snowball Sampling:

- ▶ **Description:** Participants refer other participants, often used for hard-to-reach or hidden populations.
- ▶ **Example:** Using initial participants to recruit others in a study on a niche community.

5. Self-Selection Sampling:

- ▶ **Description:** Participants volunteer to be part of the study.
- ▶ **Example:** An online survey where individuals choose to participate.

Discussion Question: Sampling Methods in computing and scientific research

Key Question:

How do probability sampling methods compare to non-probability sampling methods in terms of their advantages, disadvantages, and applicability in computing and scientific research?

► Sub-Questions to Explore:

1. What are the key differences between probability sampling and non-probability sampling?
2. How do the advantages and disadvantages of each method impact the quality and generalizability of research findings?
3. In what scenarios would probability sampling be more suitable than non-probability sampling, and vice versa?
4. How do the challenges and limitations of each sampling method affect their application in computing and scientific research?
5. Can you provide examples of how each sampling method might be applied in a real-world research study in computing or a scientific field?

► NB: This discussion question encourages a comprehensive analysis of the two sampling methods, highlighting their relative strengths and weaknesses and examining their practical applications in various research contexts

RESEARCH INSTRUMENTS

Research instruments are **tools and techniques** used to **collect data and information for a study**.

They are essential for **gathering accurate and reliable data**, which forms the basis for **analysis and conclusions** in research.

Common research instruments:

1. Surveys and Questionnaires:

- ▶ **Description:** Structured tools consisting of a series of questions designed to gather information from respondents.
- ▶ **Types:**
 - ▶ **Closed-Ended Questions:** Respondents choose from a set of predefined options.
 - ▶ **Open-Ended Questions:** Respondents provide their own answers in their own words.
- ▶ **Advantages:** Can collect data from a large number of respondents; standardized format allows for easy comparison and analysis.
- ▶ **Disadvantages:** Limited depth of responses for closed-ended questions; potential for respondent bias.

2. Interviews:

- ▶ **Description:** Direct, face-to-face, or virtual conversations between the researcher and the participant to gather detailed information.

▶ **Types:**

- **Structured Interviews:** Follow a predetermined set of questions.
- **Semi-Structured Interviews:** Combine structured questions with opportunities for open-ended responses.
- **Unstructured Interviews:** Flexible conversations with no predetermined questions, allowing for in-depth exploration.

- ▶ **Advantages:** Can provide in-depth and detailed information; allows for clarification and follow-up questions.

- ▶ **Disadvantages:** Time-consuming; potential for interviewer bias.

3. Observations:

- ▶ **Description:** The researcher observes and records behaviors, events, or conditions as they occur in their natural setting.

▶ **Types:**

- **Participant Observation:** The researcher actively participates in the setting.
- **Non-Participant Observation:** The researcher observes without involvement.

- ▶ **Advantages:** Provides real-time data; useful for studying behaviors and interactions.

- ▶ **Disadvantages:** Can be time-consuming; potential for observer bias.

RESEARCH INSTRUMENTS...cont

4. Experiments:

- ▶ **Description:** Controlled studies where variables are manipulated to observe their effects on other variables.
- ▶ **Types:**
 - ▶ **Laboratory Experiments:** Conducted in a controlled environment.
 - ▶ **Field Experiments:** Conducted in a natural setting.
- ▶ **Advantages:** Allows for control over variables; can establish cause-and-effect relationships.
- ▶ **Disadvantages:** May lack ecological validity; ethical concerns with manipulating variables.

5. Tests and Assessments:

- ▶ **Description:** Standardized tools used to measure specific skills, knowledge, or attributes.
- ▶ **Types:**
 - ▶ **Cognitive Tests:** Measure intellectual abilities.
 - ▶ **Aptitude Tests:** Assess potential to perform certain tasks.
 - ▶ **Personality Assessments:** Evaluate personality traits.
- ▶ **Advantages:** Standardized format allows for reliable and valid measurement; can be used to compare different groups.
- ▶ **Disadvantages:** May not capture the full complexity of the measured construct; potential for test anxiety.

6. Document Analysis:

- ▶ **Description:** Examination and interpretation of existing documents and records.
- ▶ **Types:**
 - ▶ **Textual Analysis:** Analyzing written content.
 - ▶ **Content Analysis:** Systematic coding and categorizing of content.
- ▶ **Advantages:** Provides historical context and background information; unobtrusive method.
- ▶ **Disadvantages:** Limited to available documents; potential for subjective interpretation.

Choosing the Right Research Instrument:

1. **Research Objectives:** Align the instrument with the goals and objectives of the study.
2. **Nature of Data:** Determine whether qualitative, quantitative, or mixed methods data are needed.
3. **Population:** Consider the characteristics of the population and their accessibility.
4. **Resources:** Assess available time, budget, and expertise.
5. **Ethical Considerations:** Ensure the instrument adheres to ethical standards and protects participant confidentiality.

DATA ANALYSIS

Data analysis is a **systematic process of inspecting, cleaning, transforming, and modeling data** with the goal of discovering useful information, drawing conclusions, and supporting decision-making.

It is a crucial step in research and involves several methods and techniques to ensure that the **data collected is accurate, reliable, and relevant to the research objectives.** 6.

Steps in Data Analysis:

1. Data Collection:

- ▶ Gather data from various sources using research instruments such as surveys, interviews, observations, experiments, etc.

2. Data Cleaning:

- ▶ Inspect and clean the data to remove any inconsistencies, errors, or missing values.
- ▶ Techniques include handling missing data, removing duplicates, and correcting inaccuracies.

3. Data Transformation:

- ▶ Convert data into a suitable format for analysis.
- ▶ Techniques include normalization, categorization, and coding.

4. Data Exploration:

- ▶ Use descriptive statistics to summarize and explore the data.
- ▶ Techniques include calculating measures of central tendency (mean, median, mode) and dispersion (standard deviation, variance).

5. Data Visualization:

- ▶ Create visual representations of the data to identify patterns, trends, and outliers.
- ▶ Tools include charts, graphs, histograms, and scatter plots.

Hypothesis Testing:

- ▶ Conduct statistical tests to determine if there is a significant relationship between variables.
- ▶ Techniques include t-tests, chi-square tests, ANOVA, regression analysis, and correlation analysis.

7. Data Modeling:

- ▶ Build models to represent the relationships between variables and make predictions.
- ▶ Techniques include linear regression, logistic regression, decision trees, and machine learning algorithms.

8. Interpretation:

- ▶ Interpret the results of the analysis to draw meaningful conclusions.
- ▶ Relate findings to the research objectives and hypothesis.

9. Reporting:

- ▶ Present the results in a clear and concise manner.
- ▶ Use tables, charts, and written summaries to communicate the findings.

Types of Data Analysis:

1. Descriptive Analysis:

- ▶ Summarizes and describes the main features of the data.
- ▶ Techniques include frequency distributions, measures of central tendency, and measures of dispersion.

2. Inferential Analysis:

- ▶ Makes inferences about the population based on sample data.
- ▶ Techniques include hypothesis testing, confidence intervals, and significance testing.

3. Predictive Analysis:

- ▶ Uses historical data to make predictions about future events.
- ▶ Techniques include regression analysis, time series analysis, and machine learning.

4. Prescriptive Analysis:

- ▶ Provides recommendations for decision-making based on data analysis.
- ▶ Techniques include optimization algorithms, simulation, and decision analysis.

Tools and Software for Data Analysis:

1. Statistical Software:

1. **SPSS:** Comprehensive statistical analysis software.
2. **R:** Open-source software for statistical computing and graphics.

3. **SAS:** Advanced analytics software for data management and analysis.

2. Spreadsheet Tools:

1. **Microsoft Excel:** Widely used for basic data analysis and visualization.
2. **Google Sheets:** Cloud-based spreadsheet tool for collaborative data analysis.

3. Data Visualization Tools:

1. **Tableau:** Powerful data visualization software for creating interactive charts and dashboards.
2. **Power BI:** Microsoft tool for business analytics and data visualization.

4. Programming Languages:

1. **Python:** Versatile language with libraries like Pandas, NumPy, and Matplotlib for data analysis and visualization.
2. **R:** Specialized language for statistical analysis and data visualization.

Importance of Data Analysis:

- 1) **Informed Decision-Making:** Provides insights and evidence to support decision-making.
- 2) **Identifying Trends:** Helps in identifying patterns and trends in the data.
- 3) **Problem Solving:** Aids in understanding and solving complex problems.
- 4) **Optimizing Processes:** Improves efficiency and effectiveness of processes and systems.
- 5) **Enhancing Research:** Strengthens the validity and reliability of research findings.

Testing Hypothesis

Hypothesis testing is a statistical method used to make inferences or draw conclusions about a population based on sample data.

It involves formulating a hypothesis and using statistical tests to determine whether there is enough evidence to reject the hypothesis.

Step-by-step guide to hypothesis testing:

1. Formulate the Hypotheses:

- ▶ **Null Hypothesis (H_0):** A statement that there is no effect or no difference. It represents the default or status quo.
 - **Example:** "There is no difference in productivity between employees who work remotely and those who work on-site."
- ▶ **Alternative Hypothesis (H_1 or H_a):** A statement that there is an effect or a difference. It represents the researcher's claim or what they aim to prove.
 - **Example:** "Employees who work remotely have higher productivity than those who work on-site."

2. Choose the Significance Level (α):

- The significance level is the probability of rejecting the null hypothesis when it is actually true. Common choices are 0.05 (5%) and 0.01 (1%).

3. Select the Appropriate Test:

- **Type of Data and Research Design:** Determine the statistical test based on the type of data (e.g., continuous, categorical) and the research design (e.g., paired samples, independent samples).

Common Tests:

- **t-test:** For comparing means between two groups.
- **Chi-Square Test:** For categorical data and testing relationships between variables.
- **ANOVA (Analysis of Variance):** For comparing means among three or more groups.
- **Regression Analysis:** For examining relationships between variables.

4. Calculate the Test Statistic:

- ▶ Use the chosen statistical test to calculate the test statistic (e.g., t-value, chi-square value).
- ▶ The test statistic measures the difference between the observed data and what is expected under the null hypothesis.

5. Determine the P-Value:

- ▶ The p-value is the probability of obtaining a test statistic at least as extreme as the one observed, assuming the null hypothesis is true.
- ▶ Compare the p-value to the significance level (α):
 - If $p\text{-value} \leq \alpha$: Reject the null hypothesis (H_0).
 - If $p\text{-value} > \alpha$: Fail to reject the null hypothesis (H_0).

Testing Hypothesis...cont

6. Make a Decision:

- ▶ **Reject H_0 :** There is sufficient evidence to support the alternative hypothesis (H_1).
- ▶ **Fail to Reject H_0 :** There is not enough evidence to support the alternative hypothesis (H_1).

7. Interpret the Results:

- ▶ **Explain the findings** in the context of the research question.
- ▶ **Discuss the implications** of the results and any potential limitations of the study.

Example:

- ▶ **Research Question:** Do employees who work remotely have higher productivity than those who work on-site?
- ▶ **Formulate Hypotheses:**
 - H_0 : There is no difference in productivity between remote and on-site employees.
 - H_1 : Remote employees have higher productivity than on-site employees.

▶ **Significance Level:**

- $\alpha = 0.05$
- ▶ **Choose the Test:**
 - Independent samples t-test (comparing means of two independent groups).
- ▶ **Calculate the Test Statistic:**
 - Compute the t-value based on the sample data.
- ▶ **Determine the P-Value:**
 - Compare the calculated p-value to the significance level (α).
- ▶ **Make a Decision:**
 - If $p\text{-value} \leq 0.05$: Reject H_0 and conclude that remote employees have higher productivity.
 - If $p\text{-value} > 0.05$: Fail to reject H_0 and conclude that there is no significant difference in productivity.
- ▶ **Interpret the Results:**
 - Provide a detailed explanation of the findings, their implications, and any limitations.

Types of errors that can be committed

- ▶ **Type I error:** it is rejecting the null hypothesis, when it is true.
- ▶ **Type II error:** It is not rejecting the null hypothesis, when it is false.

Null hypothesis	Do not reject H_0	Reject H_0
H_0 is True	Correct decision	Type I error
H_0 is false	Type II error	Correct decision

Validity and Reliability in Research

- ▶ The quality of a research study depends to a large extent on the accuracy of the data collection procedures. Reliability and validity measures the relevance and correctness of the data.

Reliability

- ▶ Reliability is the extent to which an experiment, test, or any measuring procedure yields the same result on repeated trials.
- ▶ Reliability is influenced by random error. Random error is the deviation from a true measurement due to factors that have not effectively been addressed by the researcher. As random error increases, reliability decreases. refers to the degree to which a study accurately reflects or assesses the specific concept that the researcher is attempting to measure.

Validity

- ▶ Validity It is the degree to which results obtained from the analysis of data actually represent the phenomenon under study.
- ▶ It is the accuracy and meaningfulness of inferences, which are based on the research results. It has to do with how accurately the data obtained in the study represents the variables of the study.

REPORT WRITING TECHNIQUES

- ▶ A **proposal** is a document, which details an intended activity.
- ▶ The formats for writing proposals differ from institution to institution or from department to department.
- ▶ Generally, a research proposal should include the following
 - ▶ prefatory items; Prefatory items **do not have a direct bearing on the research** itself. They assist the reader in using the research report.
 - ▶ the title page, declaration, table of contents, list of figures and tables, list of acronyms and abbreviations and an abstract.
 - ▶ Chapter one: Introduction,
 - ▶ Chapter Two: Literature review
 - ▶ Chapter Three: Methodology.
 - ▶ References, time schedule, budget and any appendices.

Final research report

- ▶ The final research report will have what is **contained in the proposal** (apart from the time schedule and budget) and
 - ▶ in addition **dedication, acknowledgement**,
 - ▶ Prefatory items: the title page, declaration, **dedication, acknowledgement** table of contents, list of figures and tables, list of acronyms and abbreviations and an abstract.
 - ▶ Chapter one: Introduction,
 - ▶ Chapter Two: Literature review
 - ▶ Chapter Three: Methodology.
 - ▶ chapter four: Data analysis and findings
 - ▶ chapter five: Summary, conclusions and recommendations.

ETHICS IN RESEARCH

- ▶ Ethics are **norms or standards** of behaviour that guide moral choices about our behaviour and our relationship with others.
- ▶ The goal of ethics in research is to ensure that **no one is harmed or suffers adverse consequences from research activities**

Ethical considerations

- ▶ Protect the rights of the participant or subject. **Benefits, Informed consent, Rights to privacy**
- ▶ Ensure the sponsor receives ethically conducted and reported research. **Confidentiality, Right to quality research and Sponsor's Ethics**
- ▶ Follow ethical standards when designing research
- ▶ Protect the safety of the researcher and team .**Safety, Ethical behaviour of assistants, Protection of anonymity**
- ▶ Ensure the research team follows the design